Views can also be materialized and refreshed concurrently. Read on to learn more.

## Improving Performance With Materialized Views

Views do not have a performance advantage when they're not materialized. They do help encapsulate complex SQL queries and can be used to create Active Record object instances backed by a view instead of a database table using Scenic. These benefits are great, but there's also another very big benefit from a performance perspective you'll work with next.

Materialized Views have great performance advantages. When application code can use Materialized Views, very high performance query code can be written because the views have pre-computed query results. Query result execution can be done concurrently as well, which you'll see how to do. This means clients can keep querying the pre-computed results while the view is recalculated in the background and replaced.

To improve performance even further, as is the case with database tables, Materialized views can have Indexes added to them. By writing queries that use Indexes on your Materialized Views you can build very high performance query code.

In the last section you worked with the SearchResult model and search_results view but this view was not materialized.

In this section you'll work with a new model called FastSearchResult.

This model was generated using the --materialized flag provided by the scenic gem that creates a Materialized View to back the model. The view was generated by running this command:

```
bin/rails g scenic:model fast_search_results --materialized
```

Both views have been added to the db/structure.sql. Search in the file for CREATE MATERIALIZED VIEW to view the definition. The db/structure.sql file shows only the definition of the view and not the precomputed query results.

Viewing the definitions of Materialized Views from psql is a bit different. Check the pg_matviews[20] System Catalog to inspect the definition:

```
SELECT * FROM pg_matviews;
```

The field ispopulated (bool) shows whether the view has been populated, and hasindexes shows whether any indexes have been added.

20. https://www.postgresql.org/docs/current/view-pg-matviews.html